

dfn: EXPERIMENTAL UNIT

→ The physical entity to which a treatment is assigned.

Eg: Participants in a clinical trial for a new vaccine.

→ There are 3 key principles of experimental Design to ensure a valid experiment: Randomisation
Replication & Blocking (later)

① dfn: RANDOMISATION:

→ Treatments should always be assigned at random to the experimental units in such a way that each unit is equally likely to receive a given treatment.
The process of assigning treatments to the experimental units in this way is called randomisation.
→ Mitigates Bias.

② dfn: REPLICATION

→ Replication has occurred when each treatment has been applied independently to more than one experimental unit.
→ Have to have to ensure valid comparisons btwn treatments. (Ensure the diff. seen is related to treatment not experimental unit variation).

dfn: COMPLETELY RANDOMISED DESIGN (CRD)

→ An experimental design where the treatments are allocated to the experimental units completely at random.
→ One independent variable / treatment